CS2014

Course Name: Computer Networks

Course Outcomes

CO1: Understand the concept of OSI and TCP/IP reference model
CO2: Understand the basics of data transmission at physical layer.
CO3: Understand the channel allocation using ALOHA, CSMA and CSMA/CD.
CO4: Apply error detection and correction technique to eliminate transmission error.
CO5: Analyze the fixed and variable length address (IPv4) subneting for the given scenarios.
CO6: Understand the design issues of the transport layer.
CO7: Understand the mechanism of protocols at application layer such as FTP, HTTP, Telnet, DNS.
CO8: Understand IPv6 addressing and differentiate it from IPv4.

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Mid Term Examination, Even Semester 2021-22

B.Tech (CSE), 2nd Year, IV Semester

BCSC0008 Computer Networks

Time: 2 Hours **Maximum Marks: 30**

Section – A 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Explain the need of OSI reference model. Discuss the role of each layer of OSI.	3	1	R	F
2	How Attenuation differ from noise. Consider a noiseless channel with a bandwidth of 3000 Hz transmitting a signal with two signal levels. What is the maximum bit rate?	3	2	U	P
3	An error correcting code has the following code words: 00000000, 00001111, 01010101, 10101010, 11110000. What is the maximum number of bit errors that can be detected and corrected? **OR** What do you understand by topology? In an organization 100 employees are working and out of which 70 an employee wants to connect each other (every employee wants to connect everyone). Suggest a suitable topology and calculate how much cable is required to connect 70 employees, if for one linking 10 m cable is required.	3	4	A	M
4	We have a pure ALOHA network with 100 stations. If T_{fr} = 1μs. what is the number of frames/s each station can send to achieve the maximum efficiency. Write vulnerable time and throughput of pure and slotted ALOHA. **OR** A network has a data transmission bandwidth of 20×10^6 bits per second. It uses CSMA/CD in the MAC layer. The maximum signal propagation time from one node to another node is 40 microseconds. What will be the minimum size of a frame in the network.	3	3	A	P

5	How do guided media differ from un-guided media? Name the advantages of optical fiber over twisted pair.	3	3	U	F

Section – B 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Consider two hosts X and Y, connected by a single direct link of rate 10^6 bits/sec. The distance between the two hosts is 10,000 km and the propagation speed along the link is 2 x 10^8 m/s. Hosts X send a file of 50,000 bytes as one large message to hosts Y continuously. Let the transmission and propagation delays be p milliseconds and q milliseconds, respectively. Then the vales of p and q are:	5	3	An	P
2	Assume that a 12-bit hamming codeword consisting of 8-bit data and 4-check bits is $d_8d_7d_6d_5c_8d_4d_3d_2c_4d_1c_2c_1$, where the databits and check bits are as follow. Databits- $d_8d_7d_6d_5d_4d_3d_2d_1$- 110x0101. Checkbits $c_8c_4c_2c_1$- y010. Find the correct values of x and y.	5	4	An	P
3	What is the difference between circuit switch and message switching? Since it is a network that uses switch, every packet goes through two links, one from source to switch and other from switch to destination. Since there are 10000 bits and packet size is 5000, two packets are sent. Transmission time for each packet is 5000 / 10^7 microseconds. Two hosts are connected via a packet switch with 10^7 bits per second links. Each link has a propagation delay of 20 microseconds. The switch begins forwarding a packet 35 microseconds after it receives the same. If 10000 bits of data are to be transmitted between the two hosts using a packet size of 5000 bits, what will be the time elapsed between the transmission of the first bit of data and the reception of the last bit of the data in microseconds? **OR** What is the need of line coding schemes? Consider a bit stream 10101010 and draw the graph for NRZ-L, NRZ-I, Manchester, and Differential Manchester Line Coding Schemes.	5	2	An	C

CO – Course Outcome, BL – Abbreviation for Bloom's Taxonomy Level (R-Remember, U-Understand, A-Apply, An-Analyze, E-Evaluate, C-Create), KL – Abbreviation for Knowledge Level (F-Factual, C-Conceptual, P-Procedural, M-Metacognitive). However, For Engg. Courses in addition to F, C, P & M include D-Fundamental Design Principles, S-Criteria and Specifications, PC-Practical Constraints, DI- Design Instrumentalities

Course Name: BMAS 1103

Course Outcome:

CO1- Compute the coefficient of correlation and regression between two variables.
CO2- Apply Chi-square test
CO3- Use Laplace transform in solving differential equations
CO4- Know the concept of analytic functions and its applications in engineering
CO5- Understand bilinear transformation and its applications
CO6- Use Cauchy's residue theorem in solving real integrals
CO7- Apply Binomial and Poisson Distributions to real-life problems

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Mid Term Examination, Even Semester 2021-22
B.Tech-II Yr., III Sem.
Engineering Mathematics III- BMAS 1103

Time: 2 Hours **Maximum Marks: 30**

Instruction for students:

1. Attempt all questions.
2. Solution must be brief, complete and to-the-point.

Section – A 3 X 5 = 15 Marks

<table>
<tr><th>No.</th><th>Detail of Question</th><th>Marks</th><th>CO</th><th>BL</th><th>KL</th></tr>
<tr><td>1</td><td>Wireless sets are manufactured with 25 solder joints each, on the average 1 joint in 500 is defective. How many sets can be expected to be free from defective joints in a consignment of 10000 sets?</td><td>3</td><td>7</td><td>An</td><td>M</td></tr>
<tr><td>2</td><td>Regression equations of two variables X and Y are: $8X - 10Y = 64$, and $40X - 18Y = 320$, What will be the mean of variables X and Y?</td><td>3</td><td>1</td><td>U</td><td>C</td></tr>
<tr><td>3</td><td>Apply χ^2 test as a test of independence to examine if there is any association between income level and type of schooling?
<table>
<tr><td>Income</td><td>Public School</td><td>Govt. School</td></tr>
<tr><td>Low</td><td>200</td><td>400</td></tr>
<tr><td>High</td><td>1000</td><td>400</td></tr>
</table>
[Given: $\chi^2 = 3.84$ at 5 % level of significance for 1 degree of freedom]</td><td>3</td><td>2</td><td>A</td><td>P</td></tr>
<tr><td>4</td><td>Estimate the parameters i.e. n and p of Binomial distribution whose mean and variance are 12 and 4 respectively.</td><td>3</td><td>7</td><td>E</td><td>F</td></tr>
<tr><td>5</td><td>If $F(t) = e^{-3t}t^3$ is an object function in time domain, Determine its image f (p) = L [F(t)] in frequency domain using Laplace transform.</td><td>3</td><td>3</td><td>E</td><td>M</td></tr>
</table>

Section – B

5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	The following data gives the aptitude test scores and productivity indices of 10 workers selected at random: Aptitude Score (X): 60 62 65 70 72 48 53 73 65 82 Productivity Index (Y): 68 60 62 80 85 40 52 62 60 81. Estimate the productivity index of a worker whose test score is 92 using regression line.	5	1	A	P
2	The demand for a particular spare part in a factory was found to vary from day to day. In a sample study, the following information was obtained: Days : Mon. Tues. Wed. Thurs. Fri. Sat. No of parts demanded : 1124 1125 1110 1120 1126 1115. Test the hypothesis at 5% level of significance using χ^2 test that the numbers of parts demanded does not depend on the day of the week. (Given that $\chi_{0.05}{}^2 = 11.07$ for 5 degree of freedom at 5% level of significance).	5	2	E	P
3	Calculate the coefficient of correlation from the following data and comment on the result: Experience (X) : 16 12 18 4 3 10 5 12 Performance (Y): 23 22 24 17 19 20 18 21.	5	1	An	F

CO – Course Outcome, BL – Abbreviation for Bloom's Taxonomy Level (R-Remember, U-Understand, A-Apply, An-Analyze, E-Evaluate, C-Create), KL – Abbreviation for Knowledge Level (F-Factual, C-Conceptual, P-Procedural, M-Metacognitive).

Course Name: Data Structures and Algorithms

Course Outcome

CO1: Understand the concepts of data structure and algorithms.

CO2: Implement of stack using array and linked list and its applications

CO3: Implement of Linked list and solve the problem of polynomials addition.

CO4: Implement of Queue using array and linked list.

CO5: Implement hierarchical based solution using tree traversal algorithms

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Mid Term Examination, Even Semester 2021-22

B.Tech (CSE/CCV/DA/IIOT/AIML/CSF), 2nd Year, 4th Semester

BCSC 0006 / Data Structures and Algorithms

Time: 2 Hours **Maximum Marks: 30**

Instruction for students: Write the complete code in one place neatly. Commenting the code is optional.

Section – A **3 X 5 = 15 Marks**

No.	Detail of Question	Marks	CO	BL	KL
1	a) What do you understand by Linked List? b) What is meant by "stack overflow" condition?	1.5 1.5	CO1	U	C
2	i) The following sequence of operation is performed on a stack: a. push(1),push(2),pop(),push(1),push(2) , pop(),pop(),pop(),push(2), pop(). b. Determine the sequence of popped out values. ii) Take a queue containing numbers 10.15,5,25,30 in which 30 has been inserted first. After performing the following operations. what would be the contents of the queue? a. Delete two elements b. Insert 7 and then 20 c. Delete an element	1.5 1.5	CO2 CO4	A	F

3	Write a program to check whether the given string is palindrome using stack. If the string is palindrome print "yes" else "no".	3	CO2	E	C
4	Define Big O notation and what is its utility in analysis of algorithms?	3	CO1	R	C
5	Suppose a linked list named myList. Write a method that inserts a node at the end of the linked list and delete the first node of the linked list. Assume that list has a tail reference as well as head reference.	3	CO3	An	F

Section – B **5 X 3 = 15 Marks**

No.	Detail of Question	Marks	CO	BL	KL
1	a) Write a recursive algorithm to find the factorial of a given number. Explain the advantages and disadvantages of using the recursive formulation compared to non-recursive formulation. b) Study the infix expression given below **a-b/(c+d)*(e-r)** convert this expression into reverse polish notation and show the stack operations for evaluating it.	2.5 2.5	CO2	C,E	C
2	What is data structure? Differentiate between linear and non-linear data structures. Explain all the operations performed on data structure.	5	CO1	U	M
3	What is queue? How is a queue differing from a Stack? Write a program to perform enqueue and dequeue operation by implementing queue using array.	5	CO4	R	P

Course Name: Introduction to Microprocessor

Course Outcome

CO1: Demonstrate the Microprocessor internal architecture and its operations.

CO2: Develop programs based on 8085 microprocessor instruction set and addressing mode.

CO3: Develop program using looping, counting, indexing, counter and time delays.

CO4: Understand the concept of stack and subroutine for modular approach.

CO5: Compare accepted standards and guidelines to select microprocessor (8085 & 8086) to meet performance requirements.

CO6: Analyze the concept of interfacing the processor to external device with I/O programming & Interrupt Driven I/O.

CO7: Understand the working of interfacing chips (8237, 8253/54, 8255 & 8259)

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Mid Term Examination, Even Semester 2021-22

B.Tech (CSE/AIML/DA/IIoT/CCV/CSF), II-Year, IV-Semester

BCSC0007: Introduction to Microprocessors

Time: 2 Hours **Maximum Marks: 30**

Instruction for students:

Attempt All Questions.

Section – A **3 X 5 = 15 Marks**

No.	Detail of Question	Marks	CO	BL	KL
1	Draw the pin diagram of 8085 microprocessor and label all the pin signals. OR Draw the internal Architecture of 8085 microprocessor and label its parts clearly.	3	1	U	C
2	What do you mean by PSW? Explain flag register of 8085 with proper format.	3	1	U,R	F,C
3	Explain the function of following instructions: a) *MOV M, D* b) *CMA* c) *ANI 64 H*	3	2	U	C
4	Define interrupt in microprocessor. Enlist various hardware interrupts on the basis of priority in 8085 microprocessor.	3	1	U	F

5	Specify the contents of Accumulator and flag register after the execution of the following program: *LXI H, 2150* *MVI A, 55* *MOV M,A* *ANI 0F* *ADD M* *HLT*	3	3	Ap	C,P

Section – B **5 X 3 = 15 Marks**

No.	Detail of Question	Marks	CO	BL	KL
1	Explain various control and status signals used in 8085 architecture.	5	1	U	C
2	Write an assembly language program to perform the following operations: (a) Load the data A248 H in BC register pair. (b) Store the data byte C7 H at memory location 2200. (c) Decrement the contents of BC register pair by one. (d) Perform AND operation of contents of memory 2200 H with the contents of B. (e) Store the result obtained in part (d) in memory 2300 H.	5	2	Ap	C,P
3	Using the software model of 8085 demonstrate ***MVI C, 54H*** instruction indicating memory address, machine cycle and T-states. Also draw the timing diagram.	5	3	Ap	C,P

Course Name: Discrete Mathematics

Course Outcome

CO1: Understand the notion of mathematical thinking and proofs to solve the problem.
CO2: Apply the basics of discrete probability and number theory to solve the real world problem.
CO3: Analyze basic discrete structures and algorithms using effectively algebraic techniques.
CO4: Analyze mathematical concepts like sets, reasoning, relational algebra and graph theory to solve optimization problems.
CO5: Analyze the validity of an argument using logical notation.
CO6: Demonstrate the basic structures of proof techniques to write and evaluate the validity of arguments.
CO7: Understand the basic principles of sets, set equalities and operations in sets.
CO8: Apply counting principles to determine probabilities.

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Mid Term Examination, Even Semester 2021-22
B.Tech.(CSE)/M.Sc(Mathematics), II Year, IV Semester
Subject Code: BCSC1010, Subject Name-Discrete Mathematics

Time: 2 Hours **Maximum Marks: 30**

Instruction for students:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section – A 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Prove that: $(A \times A) \cup (B \times C) = (A \cup B) \times (A \cup C)$	3	CO1, CO7	U	P
2	In a survey of 120 people, it was found that 65 read Newsweek magazine, 20 read both Newsweek and Time, 45 read Time, 25 read both Newsweek and Fortune, 42 read Fortune, 15 read both Time and Fortune,8 read all three magazines. Find the number of people who read at least one of the three magazines.	3	CO2	An	F
3	Prove $n^5 - n$ is divisible by 5 for all $n \in N$.	3	CO1	A	C
4	Discuss Pigeonhole Principle. There are 38 different time periods during which classes at a university can be scheduled. If there are 677 different classes. How many different rooms will be needed?	1+2	CO1, CO5	R, E	M
5	Find the solution of the recurrence relation: $a_r - 3a_{r-1} = 2.3^r$	3	CO2	E	P

Section – B 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Find whether relation R is equivalence. Let S be the set of all points in a plane. Let R be a relation such that for any two points, a and b: (a,b) ∈ R if b is within two centimeter from a.	5	CO1	A, An	C
2	Solve the differential equation $a_r - 3a_{r-1} + 2a_{r-2} = 0$, $r \geq 2$ by the generating function method with initial conditions $a_0=2$ and $a_1=3$.	5	CO2	E	P
3	How many permutations can be made with the letters of the word 'CORPORATION' and (i) In how many words vowels always come together? (ii) In how many ways consonants and vowels occurs alternatively?	1+2+2	CO8	U, E	C, M

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Mid-Term Examination, Even Semester 2021-22

B.Tech. CSE, II: Year, IV: Semester

BCSE 0901: IIoT Communication and Standard Interface

Time: 1-hour **Maximum Marks: 30**

Note: Attempt all questions. 60 x 0.5 = 30 Marks

1. The term HTTP stands for?

   A. Hyper terminal tracing program

   B. Hypertext tracing protocol

   C. Hypertext transfer protocol

   D. Hypertext transfer program

2. Flow sensor works on the principle of:

   A. Hall effect

   B. Flex counter

   C. Pulsating ratio

   D. None of the above

3. Which out of below option is not a property parameters of Thing:

   A. Logged

   B. Persistent

   C. Read Only

   D. TimeStamp

4. Wi-Fi stands for____

A. Wireless function

B. Wireless fidelity

C. Wireless functioning

D. None of above

5. ISR stands for

A. Interface service routine

B. Interrupt Service Runner

C. Interrupt Service Routine

D. Interface Standard Routine

6. Name the service used to access historical data from database.

A. QueryPropertyHistory

B. QueryRequestHistory

C. QuerySetProperty

D. None of the above

7. Select the property used to bind the Things Realtime Properties data to Mashup widgets.

A. GetProperties

B. SetProperties

C. GetsProperties

D. Get-Properties

8. Thingworx is the product of ------- company.

A. PTTC

B. PTC

C. DASSAULT

D. RENOULT

9. API stands for ..... .

A. Application programming interface

B. Android programming interface

C. Arduino protocol information

D. Application protocol interface

10. Boolean expression Y=A +B is True for

A. OR

B. AND

C. NOT

D. XOR

11. If a PROM is programmed with the error or needs Updating Then

A. It is replaced by a new one

B. We reprogram it

C. Both of the above

D. None of these

12. Which gate gives Multiplication Output if the input is A and B

A. OR

B. AND

C. NOT

D. XOR

13. Which one is Faster?

A. SRAM

B. DRAM

C. None of the above

D. Both having the same speed

14. Logic Gates Works on

A. Analog values

B. Digital value

C. Both

D. None

15. If A=1 and B=1, we are using XOR gate then Y equals to

A. 1

B. 0

C. Both

D. None

16. Primary Memory are types of:

A. RAM, ROM

B. Floppy Disk

C. Hard disk

D. None of above

17. NAND gate is made up of

A. AND, NOT

B. OR

C. NOT

D. XOR

18. Which is true for multiplexer

A. Many inputs

B. Two outputs

C. Many outputs

D. One input

19. Which of the following best describes EPROMs?

A. EPROMs can be programmed only once.

B. EPROMs can be erased by UV.

C. EPROMs can be erased by shorting all inputs to the ground.

D. All of the above

20. UART stands for:

A. Universal asynchronous receiver-transmitter

B. Uniform asynchronous receiver-transmitter

C. Unified asynchronous receiver-transmitter

D. None of the above

21. What type memory device is used in Arduino?

A. EEPROM

B. PENDRIVE

C. CD

D. MAGNETIC TAPES

22. Which among them is not an internal storage?

A. Registers

B. ROM

C. EEPROM

D. Optical disk

23. What are Thing Templets in Thingworx?

A. Set of predefined properties.

B. Holds the widgets

C. Both of the above

D. None of the above

24. Which data type is supported in Thingworx.

A. String

B. Number

C. Integer

D. All of the above

25. The command which is used to display the characters in the serial monitor is:

A. Serial

B. Serial.type

C. Serial.println

D. Serial.printing

26. Which of the following is true about IoT?

A. The term Things in the Internet of Things refers to anything and everything in day to day life.

B. IoT has greater transparency, control, and performance.

C. Both A and B.

D. None of the above.

27. What is true about memory unit?
   A. A memory unit is the collection of storage units.
   B. The memory unit stores the binary information in the form of bits.
   C. Both A and B
   D. None of the above

28. In how many categories memory/storage is classified?
   A. 2
   B. 3
   C. 4
   D. 5

29. Which of the following is correct example for Secondary Memory?
   A. Magnetic disks
   B. Tapes
   C. EEPROM.
   D. Both A and B

30. How many voltage levels are present when a pin uses digitalWrite()?
   A. 1
   B. 2
   C. 3
   D. 4

31. Can the digitalRead() function identify analog voltages?
   A. Yes, it can
   B. No, it cannot
   C. Yes, it can but only 2
   D. Yes, it can but only 4

32. What type of signal does the analogWrite() function output?

A. Pulse Width Modulated Signal
B. Pulse Code Modulated Signal
C. Pulse Amplitude Modulated Signal
D. Frequency Modulated Signal

33. What is the resolution of analog readings on the Arduino Uno?
A. 5V/1023
B. 3.3V/1024
C. 5V/2096
D. 3.3V/100

34. Difference between the = and the == operators?
A. Assignment & Comparison
B. Comparison & Assignment
C. Nor A neither B
D. Both A & B

35. Communication in UART is _____
A. Only simple
B. Only duplex
C. Only full duplex
D. Simplex, half duplex, full duplex

36. What is the full form of SRAM?
A. Synchronized Random Access Memory
B. Static Random Access Memory
C. Simplified Random Access Memory
D. Segregated Static Random Access Memory

37. What is the use of the RESET button on the Arduino UNO?
A. To restart the code present in the Arduino
B. To restore a previous version of a code
C. To erase the code present in the Arduino
D. To erase the SRAM in Arduino Microcontroller

38. What does the analogRead() function do physically when invoked in a code?

A. It accepts a digital Input
B. Reads the raw voltage value
C. Activates the ADC Circuit to read voltage values
D. Activates the DAC Circuit to read voltage values

39. Which function executes first when a sketch is built?

A. loop()
B. setup()
C. init()
D. main()

40. What is the use of #define?

A. To define macros
B. To define pointer variables
C. To define file names
D. To create variables

41. What is Mashup in Thingworx?

A. Container to design Digital Model
B. Container to design UI/UX for Visualization
C. None of the above
D. All of the above

42. Define the use of Auto-refresh widget in Thingworx?

A. To delete the previous value
B. To download mashup
C. To refresh the realtime data into mashup
D. None of the above

43. What is ValueStream in Thingworx?

A. It is used to stream data from database.
B. It is used to define properties
C. It is used to classify different properties

D. All of the above

44. Can we send data to Thingworx using RESTApi?
A. True
B. False

45. What are Labels in widgets?
A. Used to write text on Mashups.
B. To define pointer variables
C. To define file names
D. To create variables

46. Define Shapes in Widgets.
A. To add data into mashup
B. To add value to the widgets
C. To design different shapes in mashup
D. None of the above

47. What is state definition in Thingworx?
A. To define Different state with different style
B. To define data
C. Both A & B
D. None of the above.

48. What is the use of "Logged" parameter while defining property in Thingworx?
A. To enable the Property
B. To make read only
C. To store the data into database
D. None of the above

49. What is the use of "Read Only" parameter while defining property in Thingworx?
A. To enable the Property
B. To make read only so data cannot be changed

C. To update the real-time data
D. None of the above

50. Find the best option that describes Time-series chart in Thingworx Mashup.
A. It shows time stamped data.
B. It shows only real-time data
C. Both A& B
D. None of the above

51. "Data Filter" widget in Thingworx is used to:
A. Query & Filter the data in GUI Mode.
B. Delete the data
C. Both A & B
D. None of the above

52. Thingworx can provide End-to-End IOT solutions.
A. True
B. False

53. What do you understand by Responsive Mashup?
A. All the widgets will align according to User
B. All the widgets will align automatically according to distribution setting
C. Both A & B
D. None of the above

54. What do you understand by Static Mashup?
A. All the widgets will align according to User
B. All the widgets will align automatically according to distribution setting
C. Both A & B
D. None of the above

55. It starts with a /* and continues until a */ What does this do?

A. Loads a sketch
B. Makes comments
C. Compiles quicker
D. Makes stars appear

56. What rate can define the timing in the UART?

A. Bit rate
B. Baud rate
C. Speed rate
D. Voltage rate

57. Thingworx can be used design Dynamic Mahsup to display realtime data.

A. True
B. False

58. "Subscription" in Thingworx is used to trigger an output whenever any event occurs.

A. True
B. False

59. "Services" in Thingworx can be written by user?

A. True
B. False

60. "Widgets" UI in Thingworx can be modified in:

A. GUI Mode
B. CUI Mode
C. None of the above
D. Both A & B

SET-A

Course Name: English for Professional Purposes II

Course Outcomes:

CO1- Students will be able to communicate more accurately, briefly and clearly.
CO2- Students will be able to create better PowerPoint presentations.
CO3- Students will develop confidence in presentations in front of peers.
CO4- Students will be able to understand problems during verbal and non-verbal communication and rectify them.
CO5- Students will be more conscious in academic and professional communication, written and spoken.

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Mid Term Examination, Even Semester 2021-22
B.Tech. (All), 2nd Year, Semester IV
BELH0004: English for Professional Purposes II

Time: 2 Hours **Maximum Marks: 30**

Section – A

Answer any six questions. **2 X 6 = 12 Marks**

No.	Detail of Question	Marks	CO	BL	KL
1	Follow the instructions and answer the questions. a. What would be a more brief way of writing the same sentence given below? Trouble is caused when people disobey rules that have been established for the safety of all. b. What would be a clearer way to write the same sentence given below? Ayush hit the boy with a book.	2	CO1	A	C
2	Write two reasons behind analyzing locale.	2	CO 3	R	C
3	How does bad eye contact cause miscommunication? Explain with an example.	2	CO 4	C	C
4	Write one do and one don't of making a PPT for presentation.	2	CO 2	R	C
5	Suppose you are in your office and one of your colleagues calls another colleague "blind". Is the person being politically correct? If not, explain why.	2	CO 5	An	C
6	Mention 4 innovative ways of introducing a topic in a presentation.	2	CO 3	R	C
7	Write two reasons behind analyzing the audience before a presentation.	2	CO 3	R	C

8	Identify the problem with the following sentence. Rewrite it, if any. "Each student needs to submit his project by the end of this week."	2	CO 5	An	C

CO – Course Outcome, BL – Abbreviation for Bloom's Taxonomy Level (R-Remember, U-Understand, A-Apply, An-Analyze, E-Evaluate, C-Create), KL – Abbreviation for Knowledge Level (F-Factual, C-Conceptual, P-Procedural, M-Metacognitive)

Section - B

Answer all the questions. **(72 × 0.25 = 18)**

1. Choose the grammatically correct sentence.

A. If you hadn't insist on changing your shoes we wouldn't have missed the bus.
B. If you hadn't insisted on changing your shoes we wouldn't have missed the bus.
C. If you hadn't insisting on changing your shoes we wouldn't have missed the bus.
D. If you hadn't insisted on changing your shoes we wouldn't missed the bus.
E. If you hadn't insisted on changing your shoes we wouldn't have miss the bus

2. Choose the grammatically correct sentence.

A. I wish I know his sister.
B. I wish I knew his sister.
C. I wish I known his sister.
D. I wish I knowing his sister.
E. I wish I have know his sister.

3. Choose the correct option to fill in the blank.

Water ___ of hydrogen and oxygen.
A. consists
B. consist
C. consisting
D. have consist
E. is consists

4. Choose the correct option to fill in the blank.
I ___ going to school last year.
A. start
B. started
C. am starting
D. have start
E. starting

5. Choose the correct option to fill in the blank.
___ to Delhi yet?
A. Have you been
B. Has you been
C. Have you be
D. Has you gone
E. Had you

6. Choose the correct option to fill in the blank.
I ___ losing money in gambling.
A. have been
B. have
C. be
D. has been
E. will been

7. Choose the correct option to fill in the blanks.
I ___ wait till you ___ your lunch.
A. shall, finish
B. shall, are finished
C. shall, are finish
D. have, finish
E. should, finishing

8. Choose the correct option to fill in the blank.
When Kasturba died, she and Gandhi ___ for nearly fifty years.
A. had been married
B. married
C. were marry
D. marry
E. was married

9. Choose the correct option to fill in the blanks.
He ___ and ___ in his native city of Shimla.
A. live, died
B. lived, die
C. lived, died
D. is living, die
E. lived, dying

10. Choose the correct option to fill in the blank.
I am sure that he ___ a lot of friends in Delhi.
A. meet
B. will meet
C. may meet
D. might meet
E. meeting

11. Identify the option which is similar in meaning to the phrase in *italics*.
Keep them posted
A. Keep up with someone
B. Keep them waiting
C. Keep them informed about a situation
D. Keep participating
E. Keep the post

12. Identify the option which is similar in meaning to the phrase in *italics*.
Leverage (in business context)
A. To utilize something to the business's advantage
B. To utilize business advantage
C. Describes constructive comments
D. Describes something that deserves closer inspection
E. To make progress

13. Choose the correct option to fill in the blanks.
If it ___, I ___ completed the work.
A. had rained, would
B. hadn't rained, would have
C. rains, would have
D. rain, would
E. had rained, will

14. Choose the correct option to fill in the blanks.
If I ___ a lot of work to do, I always ___ for help.
A. had, look
B. will have, ask
C. have, ask
D. am having, ask
E. have, looking

15. Choose the correct option to fill in the blanks.
We ___ to the movie if the weather ___ bad.
A. should be, won't be
B. went, isn't
C. will go, isn't
D. would go, aren't
E. are going, wasn't

16. Choose the correct option to fill in the blanks.
We ___ have enough time to buy the tickets if we ___ half an hour early.
A. will, had left
B. may, left
C. will, leave
D. should, had left
E. would, leaves

17. Choose the correct option to fill in the blanks.
If they ___ that match, the club ___ fired the manager.
A. had win, will
B. hadn't, would
C. had won, would
D. hadn't won, would have
E. hadn't win, would have

18. Choose the correct option to fill in the blanks.
My brother___ to the concert if he ___ the tickets earlier this week.
A. would have gone, had bought
B. will go, had bought
C. would have, bought
D. will go, had bought
E. went, bought

19. Choose the correct option to fill in the blanks.
If the guard ___ a master key, he ___ the door for them.
A. had had, will unlock
B. had had, would have unlocked
C. have, would unlock
D. have, would locking
E. had, will unlock

20. Choose the correct option to fill in the blanks.
If they had ___ Ramesh more time, he would ___ able to do a better job.
A. give, have been
B. given, have been
C. given, been
D. giving, have been
E. would, be

21. Choose the correct option to fill in the blanks.
___ you still love me if I ___ a beard ?
A. Would, growing
B. Will, grown
C. Would, grew
D. Could, grown
E. Would, grown

22. Choose the correct option to fill in the blanks.
If you ___ yesterday's paper, you ___ the news.
A. had read, would have known
B. read, would known
C. read, will known
D. reading, know
E. had read, know

23. Choose the correct causative sentence which may follow.
The boys returned home after playing in the muddy water.
A. Their mother will make them clean the room.
B. They be made by the mother to clean the room.
C. They should be asked to clean the room.
D. They had cleaned the room.
E. They had someone cleaned the room.

24. Choose the correct causative sentence which may follow.
An old man was murdered in the hotel last night.
A. Police didn't let anyone leave the hotel.
B. Police let no one left the hotel.
C. Police do not allow to leave.
D. Police let no one has left the hotel.
E. Police want no one leave the hotel.

25. Choose the correct passive causative sentence which may follow.
My car has broken down.
A. I will get it repaired.
B. I will get it repair by the mechanic.
C. I will get someone repair it.
D. Mechanic will repair it.
E. The mechanic repaired it.

26. Choose the correct causative sentence which may follow.
Last night, my car broke down.
A. I had the mechanic fix it.
B. I have the mechanic fixed it.
C. I make it fixed by the mechanic.
D. I have mechanic fixing it.
E. I made the mechanic fixed it.

27. Choose the correct causative sentence which may follow.
I could not submit the assignment on time.
A. My teacher had me write it.
B. My teacher have asked me to write it.
C. My teacher want me write it.
D. My teacher would ask me to write it.
E. My teacher is asking me to write it.

28. Choose the correct causative sentence which may follow.
The student failed in all the subjects.
A. His teacher had the guardian called.
B. His teacher have called the guardian.
C. His teacher had called the guardian.
D. His teacher had the guardian calling.
E. His teacher had the guardian call.

29. Choose the correct causative sentence which may follow.
Bittu's room was very dirty.
A. He have the room clean.
B. He had the room clean.
C. He had the room cleaned.
D. He had the room cleaning.
E. He have the room cleaned.

30. Choose the correct passive causative sentence which may follow.
I have a severe toothache.
A. I will have it remove.
B. I got it remove.
C. I got it removed.
D. I got it removing.
E. I get it removing.

31. Choose the correct form of causatives.
Nidhi ___ her ___ her homework.
A. had, doing
B. made, does
C. make, doing
D. made, do
E. had, doing

32. Choose the correct causative sentence which may follow.
She forgot to carry cash with her.
A. I will let her money borrowed.
B. I will give her some money.
C. I will let her borrow some money.
D. I will be borrowing some money.
E. She borrowed some money.

33. Choose the correct option to fill in the blank..
My friends ___ at my house at 7 pm last night.
A. arrive
B. arriving
C. arrived
D. arrival
E. were arrived

34. Identify the idiom from the meaning given below.
Examining a group of people and deciding which are suitable and which are not
A. Separate the sheep from the goats
B. Shape up or shape out
C. Show someone the ropes
D. Work to rule
E. Get hold of

35. Choose the grammatically correct sentence.
A. If she was well, she would travel more.
B. If she was well, she travel more.
C. If she was well, she is travelling more.
D. If she was well, she travelled more.
E. If she was well, she would be travel more

36. Choose the grammatically correct sentence.
A. If it was not so late, we are visit them.
B. If it were not so late, we would visit them.
C. If it was not so late, we will visiting them.

D. If it were not so late, we visited them.
E. If it will not so late, we visit them.

37. Identify the correct alternative to the underlined part of the sentence.
Nishad waited patiently while Shobha was booking the tickets.
A. had waited
B. has waited
C. waiting
D. was waiting
E. waits

38. Identify the correct alternative to the underlined part of the sentence.
They had not delivered the documents we need.
A. had not deliver
B. not delivered
C. have not delivered
D. have not deliver
E. do not delivered

39. Identify the correct alternative to the underlined part of the sentence.
I will have been relying on my Christmas bonus to pay for the gifts I buy for my family.
A. will have relied
B. will relying
C. had rely
D. rely
E. will been relying

40. Identify the correct alternative to the underlined part of the sentence.
Miriam had arrived at 5 p.m. but Whitaker had closed the store at 4 p.m.
A. has arrived
B. arrived
C. arrives
D. is arriving
E. will arrive

41. Identify the correct alternative to the underlined part of the sentence.
Raqib was telling a story about his most recent exploits when a tree branch breaks the living room window.

A. broke
B. broken
C. breaking
D. break
E. will break

42. Identify the correct alternative to the underlined part of the sentence.
Will not the train have departed by 8 p.m.?

A. Will the train haven't departed
B. Won't the train have departed
C. Will not haven't the train departed
D. Will haven't the train departed
E. Won't have the train departed

43. Identify the correct alternative to the underlined part of the sentence.
Copernicus proved that the earth moved around the sun.

A. had moved
B. was moved
C. would move
D. moving
E. moves

44. Identify the correct alternative to the underlined part of the sentence.
Have you been tried searching the almirah?

A. Have you trying
B. Did you tried
C. Have you tried
D. Have been you tried
E. Has you been tried

45. Identify the correct alternative to the underlined part of the sentence.
This is the house where my parents live. They live here since 1991.
A. lived
B. are living
C. had lived
D. will live
E. have been living

46. Identify the correct alternative to the underlined part of the sentence.
Presently, I am going to the piano teacher only on Wednesdays.
A. go
B. went
C. have gone
D. will go
E. No correction required

47. Choose the correct option to fill in the blanks.
In truth, I am not from France. I can't speak French.
But if I ___ from France, I ___ French.
A. were, could speak
B. will be, can speak
C. been, could speak
D. were, can speak
E. was, could speak

48. Choose the correct option to fill in the blanks.
In reality, Afzal has enough money. He won't ask for loan.
But if Afzal ___ enough money, he ___ for a loan.
A. had, would ask
B. didn't have, wouldn't ask
C. didn't have, would ask
D. haven't had, would ask
E. have, would ask

49. Choose the correct option to fill in the blanks.
Mala is tired. She won't finish her work.
But if Mala ___ tired, she ___ her work.
A. will not, would finish
B. was not, would finish
C. were not, wouldn't finish
D. were, would not finish
E. None of these

50. Choose the correct option to fill in the blanks.
I didn't go to the office today, so I didn't finish the report.
If I ___ to office today, I ___ my report.
A. hadn't gone, couldn't have finished
B. go, could have finished
C. went, finished
D. didn't go, could have finished
E. None of these

51. Choose the correct option to fill in the blanks.
Yasmeen didn't tell Bhanu about her problem, so he didn't help her.
If Yasmeen ___ Bhanu about her problem, he ___ her.
A. had told, could have helped
B. hadn't told, could have helped
C. had told, wouldn't have helped
D. told, would have helped
E. told, could have helped

52. Choose the grammatically correct sentence.
A. You will feel relaxed if you had watched fish swimming in the river.
B. You will feel relaxed if you watched fish swimming in the river.
C. You will feel relaxed if you had been watched fish swimming in the river.
D. You will feel relaxed if you had been watching fish swimming in the river.
E. You will feel relaxed if you see fish swimming in the river.

53. Choose the grammatically correct sentence.
A. If you were in outer space and lit a candle, the flame would perfectly round.
B. If you were in outer space and lit a candle, the flame would be perfectly round.
C. If you were in outer space and lit a candle, the flame had been perfectly round.
D. If you were in outer space and lit a candle, the flame would have been perfectly round.
E. If you were in outer space and lit a candle, the flame had perfectly round.

54. Choose the grammatically correct sentence.
A. If you rub onion juice on your skin, insects stay away.
B. If you rub onion juice on your skin, insects stayed away.
C. If you rub onion juice on your skin, insects had stayed away.
D. If you rub onion juice on your skin, insects had been stayed away.
E. If you rub onion juice on your skin, insects have stayed away.

55. Read the sentence below and choose the grammatically correct sentence which may follow.
Sagar doesn't need a coat. He won't buy one.
A. If Sagar didn't need a coat, he would buy one.
B. If Sagar didn't need a coat, he wouldn't buy one.
C. If Sagar needed a coat, he wouldn't buy one.
D. If Sagar needed a coat, he would buy one.
E. If Sagar needed a coat, he will buy one.

56. Choose the grammatically correct sentence.
A. If I had a horse, I would ride it to school.
B. If I had had a horse, I will ride it to school.
C. If I had a horse, I will ride it to school.
D. If I had been a horse, I would ride it to school.
E. If I had been a horse, I would have ridden it to school.

57. Which of the following best expresses the idiom "speed networking"?
A. Fast internet connection
B. Responding to people on social network quickly
C. Refers to a relatively new urban trend which consists in making a potential business contact by briefly talking to a series of people at an organised event and exchanging contact details.
D. Passing information on through conversation transmitted by word of mouth
E. Refers to a process by which a message or piece of information (especially gossip, rumours or scandalous news) is passed on from one person to another, and changes along the way, so that the final version is often very different from the original because the information gets distorted or exaggerated.

58. Which of the following best expresses the idiom "worth one's salt"?
A. Someone who deserves respect because they do their job well
B. The price one has to pay to someone who sells salt
C. When, during a conflict, employees decide to do only the minimum amount of work required by the company rules, and refuse any overtime
D. When someone is waiting for an opportunity to take action, especially to replace someone else in their job or position
E. Warning someone that if they do not improve, they will be fired

59. Which of the following best expresses the idiom "paid peanuts"?
A. Peanuts which someone has already paid for
B. When someone has a very low salary
C. A humorous way of referring to someone who is an accountant or who is very good at working with numbers and calculations
D. Earning at the time of training
E. When someone walks the streets or goes from company to company, usually in search of employment

60. Which of the following best expresses the idiom "above and beyond the call of duty"?
A. When someone does something beyond what is expected from him/her
B. When someone shows sheer lack of sense of responsibility
C. When someone does something which shouldn't be done

D. When someone asks someone to do something which is beyond his/her capability
E. When someone tries to play the game Call of Duty

61. Which of the following best expresses the idiom "stool pigeon"?
A. A pigeon which has a stool
B. A stool on which pigeons sit
C. A person who acts as an informer, especially one who gives information to the police or the authorities
D. A person who takes bribe
E. A person who likes to sit like a pigeon

62. Which of the following best expresses the idiom "drop someone a line"?
A. Helping someone write a letter
B. Giving someone a lift
C. Giving someone an order
D. Writing a letter to someone
E. Firing someone

63. Which of the following best expresses the idiom "work to rule"?
A. The work a ruler of a nation has to do
B. When employees work overtime
C. When employees go on a strike
D. When employees decide to do only the minimum amount of work required by company rules, and refuse any overtime
E. When employees show off their qualifications

64. Which of the following best expresses the office jargon "*on the same page*"?
A. When employees lose their job
B. Approaching something from the same point of view with the same agreed assumptions in mind
C. When employees are divided into different groups to assign them different tasks
D. The process of selecting the next manager of a company
E. When employees fail to meet deadline

65. Identify the idiom which refers to *losing a job*.
A. Getting a golden axe
B. Getting the axe
C. Getting an axe
D. Dead wood
E. Separate the sheep from the goat

66. Identify the idiom which refers to *the position of someone about to be criticized because of the responsibilities or the position they hold.*
A. Cream of the crop
B. Dead wood
C. Duty bound
D. Firing line
E. Separate the sheep from the goat

67. Choose the correct passive causative form of the sentence written below.
The hairdresser cut my hair.
A. The hairdresser was cut by my hair.
B. The hairdresser cut my hair.
C. The hairdresser got my hair cut by me.
D. I got my hair cut by the hairdresser.
E. I have got my hair cut by the hairdresser.

68. Choose the correct passive causative form of the sentence written below.
Praful will send his car to the workshop where the mechanics will repair it.
A. Praful's car will be repaired by the mechanics in the workshop.
B. Praful's car will get repaired by the mechanics in the workshop.
C. Praful will send his car to the workshop where he will get it repaired by the mechanics.
D. Praful's car was sent to the workshop where the mechanics will repair it.
E. Praful's car was repaired by the mechanics in the workshop.

69. Choose the correct passive causative form of the sentence written below.

The engineer asked her assistant to write the letter and he did it.
A. The letter was written by the engineer's assistant.
B. The engineer got the letter written by her assistant.
C. The assistant write the engineer's letter.
D. The assistant wrote the letter of the engineer.
E. The assistant was asked by the engineer to write her letter.

70. Choose the correct causative form of the sentence written below.
The teacher asked the students to work on a project and they did it.
A. The teacher got the students work on a project.
B. The students did a project as asked by the teacher.
C. The teacher and students did a project.
D. A project was done by the students as asked by the teacher.
E. The students were asked by the teacher to work on a project.

71. Choose the correct passive causative form of the sentence written below.
My uncle hired a famous architect and she built my uncle's ten-storied mansion in Hawaii.
A. My uncle had a famous architect build his ten-storied mansion in Hawaii.
B. My uncle got his ten-storied mansion in Hawaii built by a famous architect.
C. A famous architect built my uncle's ten-storied mansion in Hawaii.
D. My uncle's ten-storied mansion in Hawaii was built by a famous architect.
E. My uncle made a ten-storied mansion in Hawaii by a famous architect.

72. Identify the idiom which refers to *the act of forcing someone to retire or give up their responsibilities*.
A. Chinese whispers
B. Plum job
C. Show someone the ropes
D. Tricks of the trade
E. Put out to pasture

L-PP B

SET-B

Course Name: English for Professional Purposes II (for, BTech & BTech-Biotech)
Course Outcomes:

CO1- Students will be able to communicate more accurately, briefly and clearly.
CO2- Students will be able to create better PowerPoint presentations.
CO3- Students will develop confidence in presentations in front of peers.
CO4- Students will be able to understand problems during verbal and non-verbal communication and rectify them.
CO5- Students will be more conscious in academic and professional communication, written and spoken.

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Mid Term Examination, Even Semester 2021-22
BTech & BTech-Biotech, 2nd Year, Semester IV
BELH0004: English for Professional Purposes II

Time: 2 Hours **Maximum Marks: 30**

Section – A

Answer any six questions. **2 X 6 = 12 Marks**

No.	Detail of Question	Marks	CO	BL	KL
1	Follow the instructions and answer the questions. a. What would be a more brief way of writing the same sentence given below? Trouble is caused when people disobey rules that have been established for the safety of all. b. What would be a clearer way to write the same sentence given below? Ayush hit the boy with a book.	2	CO1	A	C
2	Write two reasons behind analyzing locale.	2	CO 3	R	C
3	How does bad eye contact cause miscommunication? Explain with an example.	2	CO 4	C	C
4	Write one do and one don't of making a PPT for presentation.	2	CO 2	R	C
5	Suppose you are in your office and one of your colleagues calls another colleague "blind". Is the person being politically correct? If not, explain why.	2	CO 5	An	C
6	Mention 4 innovative ways of introducing a topic in a presentation.	2	CO 3	R	C
7	Write two reasons behind analyzing the audience before a presentation.	2	CO 3	R	C

8	Identify the problem with the following sentence. Rewrite it, if any. "Each student needs to submit his project by the end of this week."	2	CO 5	An	C

CO – Course Outcome, BL – Abbreviation for Bloom's Taxonomy Level (R-Remember, U-Understand, A-Apply, An-Analyze, E-Evaluate, C-Create), KL – Abbreviation for Knowledge Level (F-Factual, C-Conceptual, P-Procedural, M-Metacognitive).

Section - B

Answer all the questions. **(72 × 0.25 = 18)**

1. Choose the correct option to fill in the blanks.
He ___ and ___ in his native city of Shimla.
A. live, died
B. lived, die
C. lived, died
D. is living, die
E. lived, dying

2. Choose the correct option to fill in the blank.
I am sure that he ___ a lot of friends in Delhi.
A. meet
B. will meet
C. may meet
D. might meet
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3. Identify the option which is similar in meaning to the phrase in *italics*.
Keep them posted
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E. The students were asked by the teacher to work on a project.

19. Choose the correct passive causative form of the sentence written below.
My uncle hired a famous architect and she built my uncle's ten-storied mansion in Hawaii.
A. My uncle had a famous architect build his ten-storied mansion in Hawaii.
B. My uncle got his ten-storied mansion in Hawaii built by a famous architect.
C. A famous architect built my uncle's ten-storied mansion in Hawaii.
D. My uncle's ten-storied mansion in Hawaii was built by a famous architect.
E. My uncle made a ten-storied mansion in Hawaii by a famous architect.

20. Identify the idiom which refers to *the act of forcing someone to retire or give up their responsibilities.*

A. Chinese whispers
B. Plum job
C. Show someone the ropes
D. Tricks of the trade
E. Put out to pasture

21. Choose the grammatically correct sentence.

A. If you hadn't insist on changing your shoes we wouldn't have missed the bus.
B. If you hadn't insisted on changing your shoes we wouldn't have missed the bus.
C. If you hadn't insisting on changing your shoes we wouldn't have missed the bus.
D. If you hadn't insisted on changing your shoes we wouldn't missed the bus.
E. If you hadn't insisted on changing your shoes we wouldn't have miss the bus

22. Choose the grammatically correct sentence.

A. I wish I know his sister.
B. I wish I knew his sister.
C. I wish I known his sister.
D. I wish I knowing his sister.
E. I wish I have know his sister.

23. Choose the correct option to fill in the blank.

Water ___ of hydrogen and oxygen.

A. consists
B. consist
C. consisting
D. have consist
E. is consists

24. Choose the correct option to fill in the blank.
I ___ going to school last year.
A. start
B. started
C. am starting
D. have start
E. starting

25. Choose the correct option to fill in the blanks.
If I ___ a lot of work to do, I always ___ for help.
A. had, look
B. will have, ask
C. have, ask
D. am having, ask
E. have, looking

26. Choose the correct option to fill in the blanks.
We ___ to the movie if the weather ___ bad.
A. should be, won't be
B. went, isn't
C. will go, isn't
D. would go, aren't
E. are going, wasn't

27. Choose the correct option to fill in the blanks.
We ___ have enough time to buy the tickets if we ___ half an hour early.
A. will, had left
B. may, left
C. will, leave
D. should, had left
E. would, leaves

28. Choose the correct option to fill in the blanks.
If they ___ that match, the club ___ fired the manager.
A. had win, will
B. hadn't, would
C. had won, would
D. hadn't won, would have
E. hadn't win, would have

29. Choose the correct option to fill in the blanks.
My brother___ to the concert if he ___ the tickets earlier this week.
A. would have gone, had bought
B. will go, had bought
C. would have, bought
D. will go, had bought
E. went, bought

30. Choose the correct option to fill in the blanks.
If you ___ yesterday's paper, you ___ the news.
A. had read, would have known
B. read, would known
C. read, will known
D. reading, know
E. had read, know

31. Choose the correct causative sentence which may follow.
The boys returned home after playing in the muddy water.
A. Their mother will make them clean the room.
B. They be made by the mother to clean the room.
C. They should be asked to clean the room.
D. They had cleaned the room.
E. They had someone cleaned the room.

32. Choose the correct causative sentence which may follow.
An old man was murdered in the hotel last night.
A. Police didn't let anyone leave the hotel.
B. Police let no one left the hotel.
C. Police do not allow to leave.
D. Police let no one has left the hotel.
E. Police want no one leave the hotel.

33. Choose the correct passive causative sentence which may follow.
My car has broken down.
A. I will get it repaired.
B. I will get it repair by the mechanic.
C. I will get someone repair it.
D. Mechanic will repair it.
E. The mechanic repaired it.

34. Choose the correct causative sentence which may follow.
Last night, my car broke down.
A. I had the mechanic fix it.
B. I have the mechanic fixed it.
C. I make it fixed by the mechanic.
D. I have mechanic fixing it.
E. I made the mechanic fixed it.

35. Choose the grammatically correct sentence.
A. If she was well, she would travel more.
B. If she was well, she travel more.
C. If she was well, she is travelling more.
D. If she was well, she travelled more.
E. If she was well, she would be travel more

36. Choose the grammatically correct sentence.
A. If it was not so late, we are visit them.
B. If it were not so late, we would visit them.
C. If it was not so late, we will visiting them.

D. If it were not so late, we visited them.
E. If it will not so late, we visit them.

37. Identify the correct alternative to the underlined part of the sentence.
Nishad waited patiently while Shobha was booking the tickets.
A. had waited
B. has waited
C. waiting
D. was waiting
E. waits

38. Identify the correct alternative to the underlined part of the sentence.
They had not delivered the documents we need.
A. had not deliver
B. not delivered
C. have not delivered
D. have not deliver
E. do not delivered

39. Identify the correct alternative to the underlined part of the sentence.
I will have been relying on my Christmas bonus to pay for the gifts I buy for my family.
A. will have relied
B. will relying
C. had rely
D. rely
E. will been relying

40. Identify the correct alternative to the underlined part of the sentence.
This is the house where my parents live. They live here since 1991.
A. lived
B. are living
C. had lived
D. will live
E. have been living

41. Identify the correct alternative to the underlined part of the sentence.
Presently, I am going to the piano teacher only on Wednesdays.
A. go
B. went
C. have gone
D. will go
E. No correction required

42. Choose the correct option to fill in the blanks.
In truth, I am not from France. I can't speak French.
But if I ___ from France, I ___ French.
A. were, could speak
B. will be, can speak
C. been, could speak
D. were, can speak
E. was, could speak

43. Choose the correct option to fill in the blanks.
In reality, Afzal has enough money. He won't ask for loan.
But if Afzal ___ enough money, he ___ for a loan.
A. had, would ask
B. didn't have, wouldn't ask
C. didn't have, would ask
D. haven't had, would ask
E. have, would ask

44. Choose the correct option to fill in the blanks.
Mala is tired. She won't finish her work.
But if Mala ___ tired, she ___ her work.
A. will not, would finish
B. was not, would finish
C. were not, wouldn't finish
D. were, would not finish
E. None of these

45. Choose the correct option to fill in the blanks.
I didn't go to the office today, so I didn't finish the report.
If I ___ to office today, I ___ my report.
A. hadn't gone, couldn't have finished
B. go, could have finished
C. went, finished
D. didn't go, could have finished
E. None of these

46. Which of the following best expresses the idiom "stool pigeon"?
A. A pigeon which has a stool
B. A stool on which pigeons sit
C. A person who acts as an informer, especially one who gives information to the police or the authorities
D. A person who takes bribe
E. A person who likes to sit like a pigeon

47. Which of the following best expresses the idiom "drop someone a line"?
A. Helping someone write a letter
B. Giving someone a lift
C. Giving someone an order
D. Writing a letter to someone
E. Firing someone

48. Which of the following best expresses the idiom "work to rule"?
A. The work a ruler of a nation has to do
B. When employees work overtime
C. When employees go on a strike
D. When employees decide to do only the minimum amount of work required by company rules, and refuse any overtime
E. When employees show off their qualifications

49. Choose the correct option to fill in the blank.
___ to Delhi yet?
A. Have you been
B. Has you been
C. Have you be
D. Has you gone
E. Had you

50. Choose the correct option to fill in the blank.
I ___ losing money in gambling.
A. have been
B. have
C. be
D. has been
E. will been

51. Choose the correct option to fill in the blanks.
I ___ wait till you ___ your lunch.
A. shall, finish
B. shall, are finished
C. shall, are finish
D. have, finish
E. should, finishing

52. Choose the correct option to fill in the blank.
When Kasturba died, she and Gandhi ___ for nearly fifty years.
A. had been married
B. married
C. were marry
D. marry
E. was married

53. Identify the option which is similar in meaning to the phrase in *italics*.
Leverage (in business context)
A. To utilize something to the business's advantage
B. To utilize business advantage
C. Describes constructive comments
D. Describes something that deserves closer inspection
E. To make progress

54. Choose the correct option to fill in the blanks.
If it ___, I ___ completed the work.
A. had rained, would
B. hadn't rained, would have
C. rains, would have
D. rain, would
E. had rained, will

55. Choose the correct option to fill in the blanks.
If the guard ___ a master key, he ___ the door for them.
A. had had, will unlock
B. had had, would have unlocked
C. have, would unlock
D. have, would locking
E. had, will unlock

56. Choose the correct option to fill in the blanks.
If they had ___ Ramesh more time, he would ___ able to do a better job.
A. give, have been
B. given, have been
C. given, been
D. giving, have been
E. would, be

57. Choose the correct passive causative sentence which may follow.
I have a severe toothache.
A. I will have it remove.
B. I got it remove.
C. I got it removed.
D. I got it removing.
E. I get it removing.

58. Choose the correct form of causatives.
Nidhi ___ her ___ her homework.
A. had, doing
B. made, does
C. make, doing
D. made, do
E. had, doing

59. Choose the correct causative sentence which may follow.
She forgot to carry cash with her.
A. I will let her money borrowed.
B. I will give her some money.
C. I will let her borrow some money.
D. I will be borrowing some money.
E. She borrowed some money.

60. Choose the correct option to fill in the blank..
My friends ___ at my house at 7 pm last night.
A. arrive
B. arriving
C. arrived
D. arrival
E. were arrived

61. Identify the idiom from the meaning given below.
Examining a group of people and deciding which are suitable and which are not
A. Separate the sheep from the goats
B. Shape up or shape out
C. Show someone the ropes
D. Work to rule
E. Get hold of

62. Identify the correct alternative to the underlined part of the sentence.
Raqib was telling a story about his most recent exploits when a tree branch breaks the living room window.
A. broke
B. broken
C. breaking
D. break
E. will break

63. Identify the correct alternative to the underlined part of the sentence.
Will not the train have departed by 8 p.m.?
A. Will the train haven't departed
B. Won't the train have departed
C. Will not haven't the train departed
D. Will haven't the train departed
E. Won't have the train departed

64. Identify the correct alternative to the underlined part of the sentence.
Copernicus proved that the earth moved around the sun.
A. had moved
B. was moved
C. would move
D. moving
E. moves

65. Identify the correct alternative to the underlined part of the sentence.
Have you been tried searching the almirah?
A. Have you trying
B. Did you tried
C. Have you tried
D. Have been you tried
E. Has you been tried

66. Choose the correct option to fill in the blanks.
Yasmeen didn't tell Bhanu about her problem, so he didn't help her.
If Yasmeen ___ Bhanu about her problem, he ___ her.
A. had told, could have helped
B. hadn't told, could have helped
C. had told, wouldn't have helped
D. told, would have helped
E. told, could have helped

67. Choose the grammatically correct sentence.
A. You will feel relaxed if you had watched fish swimming in the river.
B. You will feel relaxed if you watched fish swimming in the river.
C. You will feel relaxed if you had been watched fish swimming in the river.
D. You will feel relaxed if you had been watching fish swimming in the river.
E. You will feel relaxed if you see fish swimming in the river.

68. Choose the grammatically correct sentence.
A. If you were in outer space and lit a candle, the flame would perfectly round.
B. If you were in outer space and lit a candle, the flame would be perfectly round.
C. If you were in outer space and lit a candle, the flame had been perfectly round.
D. If you were in outer space and lit a candle, the flame would have been perfectly round.
E. If you were in outer space and lit a candle, the flame had perfectly round.

69. Choose the grammatically correct sentence.

A. If you rub onion juice on your skin, insects stay away.
B. If you rub onion juice on your skin, insects stayed away.
C. If you rub onion juice on your skin, insects had stayed away.
D. If you rub onion juice on your skin, insects had been stayed away.
E. If you rub onion juice on your skin, insects have stayed away.

70. Which of the following best expresses the office jargon "*on the same page*"?

A. When employees lose their job
B. Approaching something from the same point of view with the same agreed assumptions in mind
C. When employees are divided into different groups to assign them different tasks
D. The process of selecting the next manager of a company
E. When employees fail to meet deadline

71. Identify the idiom which refers to *losing a job*.

A. Getting a golden axe
B. Getting the axe
C. Getting an axe
D. Dead wood
E. Separate the sheep from the goat

72. Identify the idiom which refers to *the position of someone about to be criticized because of the responsibilities or the position they hold.*

A. Cream of the crop
B. Dead wood
C. Duty bound
D. Firing line
E. Separate the sheep from the goat

Course Name: Database Management System

Course Outcome

CO1: Understand the concept of database management systems and Relational database.
CO2: Identify the various data model used in database design.
CO3: Design conceptual models of a database using ER modeling for real life applications and construct queries in Relational Algebra.
CO4: Create and populate a RDBMS for a real-life application, with constraints and keys using RA.
CO5: Select the information from a database by formulating complex queries in RA.
CO6: Analyze the existing design of a database schema and apply concepts of normalization to design an optimal database.
CO7: Discuss indexing mechanisms for efficient retrieval of information from a database.
CO8: Discuss recovery system and be familiar with introduction to web database, distributed databases.

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Mid Term Examination, Even Semester 2021-22
B.Tech (CSE), II, IV
Subject Code & Subject Name- BCSC0003 Database Management System

Time: 2 Hours **Maximum Marks: 30**

Instruction for students:
1. Attempt all questions
2. don't switch between the section A and Section B. first solve one section then move to second
3. Use only blue pen to answer
4. No B sheet will be given

Section – A 3 X 5 = 15 Marks

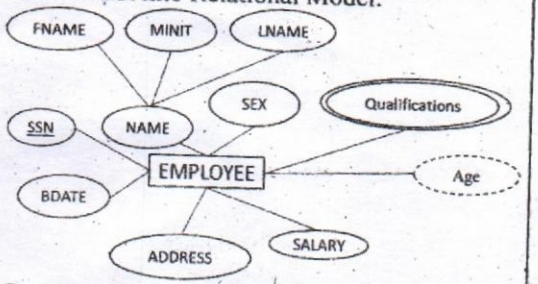
No.	Detail of Question	Marks	CO	BL	KL
1	Rohan had shifted his website where he sells caps and hats from PHP to more modern REACT.JS. He need not to make any changes in his dbms design and schema while making this shift. What concept of DBMS does he adhere to, so that no changes were required in schema while making the shift? Also, discuss types of this concept with an example.	3	CO1	An	D
2	DBMS is preferred over generic file systems. True or False with proper reasons and scenarios.	3	CO1	U	C
3	What are the various set operations in Relational Algebra? Can we apply set operations on the following relations? StudentBTech: sID | sName | CPI | Branch 1 | Amit | 2.3 | CS 2 | Ajay | 4.1 | EC 3 | Ram | 3.8 | CS	3	CO4	E	M

	StudentMBA: sID | sName | CPI 1 | Amanda | 6.1 4 | John | 2.3 9 | Samantha | 7.1 Provide a reason for your answer. If your answer is no, what changes are required to make it fit for set operations?				
4	Consider the following relations for a database that keeps track of student enrollment in courses and the books adopted for each course: STUDENT(SSN, Name, Major, Bdate) COURSE(Course#, Cname, Dept) ENROLL(SSN, Course#, Quarter, Grade) BOOK_ADOPTION(Course#, Quarter, Book_ISBN) TEXT(Book_ISBN, Book_Title, Publisher, Author) Draw a relational schema diagram specifying the foreign keys for this schema.	3	CO3	C	M
5	Differentiate among the following: a. Schema and instance b. DDL and DML c. Natural Join and Theta Join	3	CO2	R	F

Section – B

5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Discuss three schema architecture with help of diagram. Explain type of Database users.	5	CO3	R	F
2	Consider the following schema: Suppliers(*sid*: integer, sname: string, address: string) Parts(*pid*: integer, pname: string, color: string) Catalog(*sid*: integer, *pid*: integer, cost: float) a. Find the *names of suppliers* who live in "Mathura." b. Find the *details* of parts whose color is red.	5	CO5	C	M

	c. Find the *sids* of the supplier who supplies part having cost greater than 10. d. Find the *sids* of suppliers who supply some red parts. e. Find the *name of suppliers* who supply some red parts.				
3	a. Resolve ER into Relational Model:  b. Create an enhanced ER Diagram for the following scenario: Imagine you have entity *cuisine* while building the database for a restaurant. Since the menu had too many dishes, you decide to divide them into multiple *cuisines* such as *Chinese*, *Italian*, *South Indian* and many more (add as per your creativity). Draw a diagram for the above and determine specialization constraints such as partial versus total, and disjoint versus overlapping [explain reasons for your constraints choices].	5	CO2	C	M

Course Name: Operating Systems

Course Outcome

CO1- Understand the classification of operating system environment.
CO2- Understand the basic of process management.
CO3- Apply the concept of CPU process scheduling for the given scenarios.
CO4- Illustrate theprocess synchronization and concurrency process in operating system.
CO5- Analyze the occurrence of deadlock in operating system.
CO6- Describe and analyze the memory management and its allocation policies.
CO7- Understand the concepts of disk scheduling

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Mid Term Examination, Even Semester 2021-22
B. Tech. (CSE), II Year, IV Semester
Operating Systems (BCSC0004)

Time: 2 Hours **Maximum Marks: 30**

Note: Attempt all questions.

Section – A 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Consider the following set of processes that need to be scheduled on a single CPU. All the times are given in milliseconds. Process Name | Execution Time A | 6 B | 2 C | 4 D | 6 E | 3 Calculate the average process turnaround time (in msec) using the shortest job first scheduling algorithm?	3	3	A	P
2	Consider the following two-process synchronization solution. Process 0 --- Process 1 ..... Entry: loop while (turn == 1); (critical section) Exit: turn = 1; Entry: loop while (turn == 0); (critical section) Exit: turn = 0; The shared variable turn is initialized to 0 or 1. Is this solution satisfying all critical section requirement? Justify.	3	4	E	M

3	Identify and explain the type of operating systemsthat focuses on 1) CPU utilization 2) Good response time.	3	1	R,U	F
4	Differentiate between monolithic and micro kernel systems.	3	1	An	C
5	What acts as a repository of a process? What information are stored in it related to a process?	3	2	U	M

Section – B 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Discuss various process states (7 states) and their meanings with a suitable diagram.	5	2	U	F
2	Discuss Peterson's algorithm with its merits and demerits.	5	4	U, A	C
3	Consider the set of processes with arrival time (in milliseconds), CPU burst time (in milliseconds), and priority (Assume 4 is the highest priority) shown below. None of the processes have I/O burst time. Calculate the average waiting time (in milliseconds) of all the processes using pre-emptive priority scheduling algorithm.	5	3	A	P

Process	Arrival time	Burst Time	Priority
P1	0	11	2
P2	5	28	0
P3	12	2	3
P4	2	10	1
P5	9	16	4

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Mid Term Examination, Even Semester 2021-22
B.TECH. (CSE-CCV), II Year, IV Semester

Cloud & Business Process Management (BCSE0508)

Time: 2 Hours **Maximum Marks: 30**

Attempt all the questions.

Section – A 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Which three services are available in the Middleware container of ANEKA cloud architecture? Explain in detail.	3	1	R	F
2	Who are the key players of the platform as a service layer in Cloud Computing? Write the functionality of On-demand self-service with the help of an appropriate example.	3	2	An	C
3	What is a "Unified layer" in Cloud Architecture? Explain with the help of diagrammatic presentation of cloud architecture along with the functionality of virtualization.	3	4	R	F
4	Why BPM is important? Explain Business Agility in detail.	3	3	U	P
5	What is BPM? Explain with the help of its life cycle.	3	1	R	F

Section – B 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Why is cloud management essential based on the many characteristics of cloud computing? Explain with the help of a diagrammatic presentation.	5	1	U	C

2	The BPM evaluation phase makes use of available information to assess and clarify this statement in depth. Also, include the stakeholders and administrators in your response. What are the various types of business processes?	5	4	A	P
3	What is Cloud Provisioning, and how does it work? Explore several types of provisioning models. Write short notes on orchestration and automate.	5	1	R	M

Course Name: Applied Statistical Analysis

Course Outcome

CO1-Understand the different variables measurement and scalling.
CO2-Analyze the different central tendency operations.
CO3-Demonstrate the different ways to represent the data.
CO4-Analyze the hypothesis testing using different estimation and distribution methods.
CO5-Analyze the correlation and regression for the real world problem data.

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Mid Term Examination, Even Semester 2021-22
B.Tech (CSE(DA), II[nd] Year, IV Semester
Subject Code & Subject Name- BCSE0552 & Applied Statistical Analysis

Time: 2 Hours **Maximum Marks: 30**

Instruction for students:
1. Write your Roll No. on the top immediately on receipt of this question paper
2. Attempt all the questions.

Section – A 3 X 5 = 15 Marks

No	Detail of Question	Marks	CO	BL	KL
1	Differentiate among cross-sectional, time series and panel data in tabular form by specifying attribute.	3	1	U	F
2	Find the mode for the following distribution: Class - interval: 0-10 10-20 20-30 30-40 40-50 50-60 60-70 70-80 Frequency 5 8 7 12 28 20 10 10	3	2	E	C
3	Wire manufactured by a company is tested for strength. The test gives a correct positive result with a probability of 0.85 when the wire is strong, but gives an incorrect positive result (false positive) with a probability of 0.04 when in fact the wire is not strong. If 98% of the wires are strong, and a wire chosen at random fails the test, what is the probability it really is not strong enough?	3	2	A	C
4	At the Fairtown High Court, people on trial are judged as follows: • For people who really are guilty, the judgment says "Yes" 97% of the time • For people who are in fact innocent, the judgment says "Yes" 4% of the time ("false positive") If 72.5% of people on trial really are guilty, and the	3	3	A	C

	judgment for a randomly selected person says "Innocent", what are the chances that the person really is innocent?				
5	A teacher asked the students to complete 60 pages of a record note book. Eight students have completed only 32, 35, 37, 30, 33, 36, 35 and 37 pages. Find the standard deviation of the pages yet to be completed by them.	3	2	E	C

Section – B 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Suppose 1845 teenage girls are to be surveyed by a cell phone company that wants to design a new cell phone that they can successfully market to females under 20 years old. The questionnaire will likely include questions related to: age, birth date, race, area code where they live, number of texts sent per month, amount of money willing to spend per month, services they want offered, features they want included, length of time they have had a cell phone, favorite colors, etc. All of these are variables, because they will vary from individual to individual. However, only some of these variables are numerical. Identify the individuals and the numerical variables.	5	1	U	F
2	Determine the outliers of the data set. Also, evaluate the mean of the data set including the outliers and excluding the outliers. 35, 75, 20, 25, 15, 30, 30, 15, 45, 40, 110	5	3	E	P
3	Suppose a student can either submit only their SAT score or their ACT score to a particular college. Suppose their SAT score was 620 and that the SAT has a mean of 500 and a standard deviation of 100. Suppose also that the same student scored a 25 of their ACT exam and that the ACT exam has a mean of 18 and a standard deviation of 6. Which score should the student submit?	5	2	A	C

Course Name: IT Network Security (BCSE0602)

Course Outcome

CO1- Understand the fundamentals of network security
CO2- Demonstrate the principle of network security controls, protocols and devices
CO3- Understand the basic concepts of physical and host security, and secure IDS configuration & Management
CO4- Explain the network security by using VPN and wireless technologies
CO5- Explain the concepts of network traffic monitoring and analysis
CO6- Conceptualize the risk management framework and vulnerability management
CO7- Explain the concepts of data backup, recovery tools and incident response management

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Mid Term Examination, Even Semester 2021-22
B.Tech(CSF), II Year, IV Semester
IT Network Security(BCSE0602)

Time: 2 Hours **Maximum Marks: 30**

Instruction for students:
Give appropriate representation and figure to support your answer.

Section – A 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	What do you understand by network topology? Which topology is used the most for connecting computers nowadays. Differentiate switch and hub.	3	1	R	C
2	Explain the different ways of providing web security? Elaborate in detail the different protocols related to Secure Socket Layer.	3	2	U	C
3	Discuss is brief the different layers and protocols of TCP/IP protocol stack.	3	1	R	F
4	What do you mean by reconnaissance attacks? Differentiate DHCP spoofing and DHCP starvation. Also differentiate DNS poisoning and DNS cache poisoning.	3	2	R	C
5	Discuss the three important ways which can be used to provide security to a web server.	3	3	An	M

Section – B 5 X 3 = 15 Marks

No	Detail of Question	Marks	CO	BL	KL
1	What do you mean by multicasting? A company is granted the site address 156.66.0.0. The company needs 1800 subnets. Show the complete process of designing the subnets in the given scenario.	5	1	A	P
2	Show the complete process of Digital Signature for message verification. Which encryption technique is used in Digital Signature? Which is the best way for distributing public keys over the network? Explain.	5	2	A	P
3	What do you mean by physical and host security? Consider a scenario in which we need to provide the physical and host security for any organization, then what are the important steps one should consider separately for achieving physical security and host security.	5	3	An	M

Course Name: Algorithm for Intelligent Systems and Robotics

Course Outcome:

CO1- Understand the Theoretical aspect of various searching strategies to solve AI Problems.

CO2- Under a range of techniques of intelligent systems across artificial intelligence and intelligent Agents.

CO3- Describe and Apply various techniques for logic programming and Machine Learning.

CO4-Understand the Practical Implementation of various tools for the implementation of Robotics.

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Mid Term Examination, Even Semester 2021-22
Program- B. Tech, II Year, IV Semester
Subject Code & Subject Name- Algorithm for Intelligent Systems and Robotics
BCSE0702

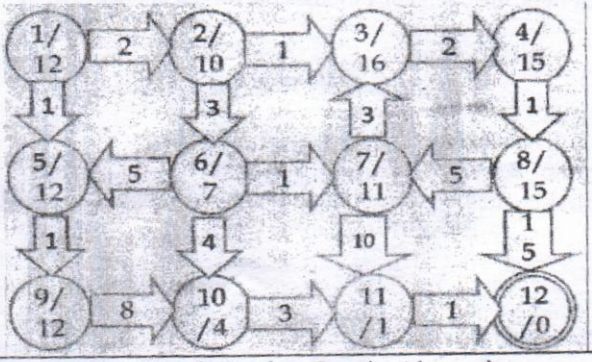
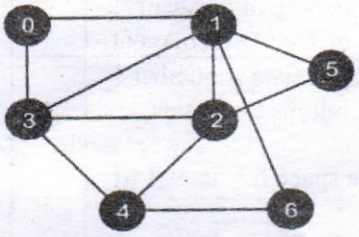
Time: 2 Hours **Maximum Marks: 30**

Instruction for students: Attempt all Questions

Section – A 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	What are the factors on which rationality of an agent depends? Give PEAS description of Taxi Driver Agent.	3	CO2	U	F
2	Consider a state space where the start state is number 1 and successor function for state n returns two states, number 2n and 2n+1. a. Draw the portion of the state space for states 1 to 15. b. Suppose the goal state is 11. List the order in which nodes will be visited using breadth first search.	3	CO1	A	C
3	What is Admissible Heuristic. Design an Admissible heuristic for 8 puzzle Problem.	3	CO2	U	F
4	Explain 8 Queen Problem. Write a Pseudo code to design a solution for it.	3	CO3	A	P
5	Differentiate the following a. Fully observable vs Partially observable task Environment b. Single vs Multiple Agents task Environment	3	CO2	U	F

Section – B 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Describe the Steepest Ascent Hill Climbing Search Technique. What are the Potential Problems associated with it and its corresponding Solution? Explain	5	CO1	U	F
2	Apply A* Search to the following graph to find the minimum cost path from node 1 to node 12 	5	CO1	E	P
3	Apply DFS on the following Graph. Also write a recursive function for it. 	5	CO1	A	P

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Mid-Term Examination, Even Semester 2021-22

B.Tech. CSE (IIoT) & B.Tech CSE, II & IV : Year, IV & VIII : Semester

BCSE 0903: Data Sciences and Analytics

Time: 1-hours **Maximum Marks: 30**

Note: Attempt all questions. 60 x 0.5 = 30 Marks

1. Pandas is an open-source _____ Library?

   A. Ruby

   B. Javascript

   C. Java

   D. Python

2. Pandas key data structure is called?

   A. Keyframe

   B. DataFrame

   C. Statistics

   D. Econometrics

3. In pandas, Index values must be?

   A. unique

   B. hashable

   C. Both A and B

   D. None of the above

4. Which of the following is correct Features of DataFrame?

A. Potentially columns are of different types

B. Can Perform Arithmetic operations on rows and columns

C. Labeled axes (rows and columns)

D. All of the above

5. A PANDA DataFrame is a ___ container of data.

A. 1D

B. 2D

C. 3D

D. Infinite

6. Which of the following is true?

A. If data is an ndarray, index must be the same length as data.

B. Series is a one-dimensional labeled array capable of holding any data type.

C. Both A and B

D. None of the above

7. Which of the following takes a dict of dicts or a dict of array-like sequences and returns a DataFrame?

A. DataFrame.from_items

B. DataFrame.from_records

C. DataFrame.from_dict

D. All of the above

8. What will be output for the following code?

```
import pandas as pd
import numpy as np
s = pd.Series(np.random.randn(4))
print s.ndim
```

A. 0

B. 1

C. 2

D. 3

9. Which of the following indexing capabilities is used as a concise means of selecting data from a pandas object?

A. In

B. ix

C. ipy

D. iy

10. Which of the following libraries allows to manipulate, transform and visualize data easily and efficiently.

A. Pandas

B. NumPy

C. Matplotlib

D. All of the above

11. The Process of describing the data that is huge and complex to store and process is known as:

A. Analytics

B. Data mining

C. Big data

D. Data warehouse

12. _____ is an important library used for analyzing data.

A. Math

B. Random

C. Pandas

D. None of the above

13. Important data structure of pandas is/are _____

A. Series

B. DataFrame

C. Both of the above

D. None of the above

14. Pandas Series can have _____ data types.

A. float

B. integer

C. String

D. All of the above

15. Which of the following command is used to install pandas?

A. pip install pandas

B. install pandas

C. pip pandas

D. None of the above

16. Which of the following command is used to import pandas?

A. import pandas as pd

B. import pandas

C. from pandas import *

D. All of above

17. A Series by default have numeric data labels starting from _____

A. 3

B. 2

C. 1

D. 0

18. The data label associated with a particular value of Series is called its_____

A. Data value

B. Index

C. Value

D. None of the above

19. Which of the following function/method help to create Series?

A. series( )

B. Series( )

C. createSeries( )

D. None of the above

20. Write the output of the following:

```
>>> import pandas as pd
>>> series1 = pd.Series([10,20,30])
>>> print(series1)
```

A. Output:

a. 0 10

b. 1 20

c. 2 30

d. dtype: int64

B. Output:

a. 10

b. 20

c. 30

d. dtype: int64

C. Output:

a. 0

b. 1

c. 2

d. dtype: int64

D. None of the above

21. When you print/display any series then the left most column is showing _____ value.

A. Index

B. Data

C. Value

D. None of the above

22. How many values will be there in array1, if given code is not returning any error?

>>> series4 = pd.Series(array1, index = ["Jan", "Feb", "Mar", "Apr"])

A. 1

B. 2

C. 3

D. 4

23 Which of the following statement will create an empty series named "S1"?

A. S1 = pd.Series(None)

B. S1 = pd.Series( )

C. Both of the above

D. None of the above

24. How many elements will be there in the series named "S1"?

```
>>> S1 = pd.Series(range(5))
>>> print(S1)
```

A. 5

B. 4

C. 6

D. None of the above

25. When we create a series from dictionary then the keys of dictionary become _____

A. Index of the series

B. Value of the series

C. Caption of the series

D. None of the series

26. Write the output of the following:

```
>>> S1=pd.Series(14, index = ['a', 'b', 'c'])
>>> print(S1)
```

A.

a. a 14

b. b 14

c. c 14

d. dtype: int64

B.

a. a 14

b. dtype: int64

C. Error

D. None of the above

27. Write the output of the following:

```
>>> S1=pd.Series(14, 7, index = ['a', 'b', 'c'])
>>> print(S1)
```

A.

a. a 14
b. b 7
c. c 7
d. dtype: int64

B.

a. a 14
b. b 7
c. dtype: int64

C. Error

D. None of the above

28. Which of the following code will generate the following output?

```
Jan 31
Feb 28
Mar 31
dtype: int64
```

A. import pandas as pd

a. S1 = pd.Series(data = [31,28,31], index=["Jan","Feb","Mar"])

b. print(S1)

B. import pandas as pd

a. S1 = pd.Series([31,28,31], index=["Jan","Feb","Mar"])

b. print(S1)

C. Both of the above

D. None of the above

29. What type of error is returned by following code?

```
import pandas as pd
S1 = pd.Series(data = (31, 2, -6), index = [7, 9, 3, 2])
print(S1)
```

A. SyntaxError

B. IndexError

C. ValueError

D. None of the above

30. PANDAS stands for _____.

A. Python Data Analysis

B. Panel Data analyst

C. Panel Data

D. Panel Dashboard

31. We can imagine a Pandas DataFrame as a _____ in a spreadsheet

A. Column

B. Cell

C. Table

D. None of the above

32. Which of the following statement will print Series 'S1' in reverse order?

A. print(S1[: : 1]

B. print(S1[: : -1]

C. print(S1[-1: : 1]

D. print(S1.reverse( ))

33. Select the command to display both row and column index label of dataframe 'exp'.

A. print(exp.index)

B. print(exp.index())

C. print(exp.axes())

D. print(exp.axes)

34. To access value of dataframe using column label we can use:

A. loc

B. <dataframe object>.<column label>

C. Both a & b

D. None

35. What will be output of following code:

```
import pandas as pd
L = [so, no, go, do, to])
df = pd.DataFrame(L)
print(df[-2:]
```

A. 2 go
   1 no
   0 so

B. 3 do
   4 to

C. 4 to
   3 do

D. 0 so
   1 no
   2 go

36. To access individual item from a dataframe 'df' which of the following is not correct:

A. df.at[row_index, column_index]

B. df.iat[row_index, column_index]

C. df.<column label> [row_index]

D. df.iloc[row_index, column_index]

37. Which of the following is not an attribute of dataframe:

A. axes

B. empty

C. transpose

D. Size

38. To count total no of elements in a DataFrame we use :

A. size

B. len

C. count

D. values

39. A dataframe object can be created using:

A. Python Dictionary

B. Python List

C. Panda Series

D. All

40. What will be output of

print(df.loc[:])

A. Display 'Error'

B. Display all rows

C. Display all columns

D. Display all rows and columns

41. Which of the following can be used to add a new column in existing dataframe:

A. loc

B. <dataframe_object>.<column_lable>

C. at

D. iloc

42. A DataFrame has two axes, where axes = 0 represents:

A. Row

B. Column

C. Index

D. Values

43. To count total no of rows of DataFrame 'df' which of the following command can be written:

A. len(df)

B. df.len

C. df.len()

D. All are correct

44. Aryan wants to access 'Author' column from 'Book' dataframe. Suggest him to correct statement:

A. print(Book.Author)

B. print(Book['Author'])

C. Both a & b are true

D. (a) True (b) False

45. If a dimension is given as ____ in a reshaping operation, the other dimensions are automatically calculated.

A. Zero

B. One

C. -1

D. Infinite

46. What the following statement will display?

df.iloc[2 : 7, : 3]

A. Display 2nd to 7th row and first three columns.

B. Display 3rd to 7th row and first three columns.

C. Display 2nd to 7th row and first four columns.

D. Display 3rd to 7th row and first four columns.

47. What will be output of the following code?

```
list1 = [[10,20], [30,40], [50,60]]
df = pd.DataFrame(list1)
print(df)
```

A.

0	10	20
1	30	40
	50	60

B.

10	0	1
	30	50

20 40 0

C.

	0	1
0	10	20
1	30	40
2	50	60

D.

	0	1	2
0	10	30	50
1	20	40	60

48. Which of the following is Missing data?

A. NULL

B. NaN

C. None

D. All Above

49. To access individual item from dataframe 'df' which of the following is Not correct?

A. df.loc[2,2]

B. df.iat[2,2]

C. df.at[2,2]

D. df[0,0]

50. Which of the following is not a iterative function for dataframe?

A. iterrows()
B. iteritems()

C. itercolumns()
D. none

51. For given DataFrame 'df'

	A	B
0	15	18
1	20	25
2	40	50
3	70	44

What will be output of the following code?

df.tail(2)

A.

	A	B
3	70	44
2	40	50

B.

	A	B
2	40	50
3	70	44

C.

	A	B
1	20	25
0	15	18

E. None

52. Which of the following statement is incorrect for adding new column 'sal' in dataframe emp?

A. emp.sal[200,300]

B. emp.loc[:,'sal']=[200,300]

C. emp = emp.assign('sal'= [200,300])

D. emp.at[:, 'sal'] = [200,300]

53. Which of the following statement is not true?

A. When values are assigned to column of dataframe for non-existing columns, it will add a new column.

B. In iloc, like slices end index/position is excluded when given as start: end

C. While creating a dataframe from 2D dictionary, keys of all inner dictionaries must be same in number or name.

D. By default in drop function of dataframe object value of axis is 1 always.

54. A head() will display

A. First five rows of dataframe

B. All rows of a dataframe

C. First row of dataframe

D. Nothing will be displayed

55. Astha has given following command to delete a column 'pub' from 'book' dataframe.

book.drop(pub)

But she is not getting desire output, help her to select correct statement.

A. book.drop(rpubl)

B. book.drop(['pub'], axis = 0)

C. book.drop(rpubl, axis = 1)

D. none

56. A _____ is a collection of data values and operations that can be applied to that data.

A. Data Structure

B. Data Frame

C. Table

D. None of the above

57. Which of the following statement is wrong?

A. We can create Series from Dictionary in Python.

B. Keys of dictionary become index of the series.

C. Order of indexes created from Keys may not be in the same order as typed in dictionary.

D. All are correct

58. to count non NaN values of dataframe, we can use

A. size

B. len

C. count

D. values

59. Which of the following is used to merge two dataframes?

A. Join()

B. Merge()

C. Extend()

D. Append()

60. Which of the following is used to fill missing values in a dataframe?

A. skipna

B. fillna

C. replacena

D. inplacena